9-1 Database [1]

Point!

1 Database Management System

(1) ([1]**Database**) : An organized collection of data, organized and stored in a format that makes it easy to access for specific purposes. In addition to collecting data, it also makes it easy to search, process, and share.
<Examples> Smartphone contact lists, company customer information, etc.

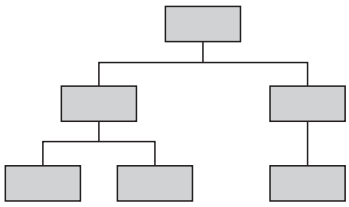
(2) ([2]**Database management system (DBMS)**) : A system that creates, operates, and manages databases.

(3) Database Management System Functions
[1] Data ([3]**consistency**) : Ensuring that concurrent operations for shared data do not cause inconsistencies.
[2] Data ([4]**integrity**) : Preventing duplication, tampering, and unauthorized registration or updating of data.
[3] Data ([5]**independence**) : Managing databases separately from the programs that use them.
[4] Data ([6]**confidentiality**) : Setting access permissions and performing authentication.
[5] Data ([7]**availability**) : Performing backups, restoration, and recovery to prepare for possible failures so that data can be recovered.

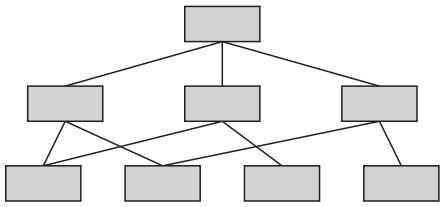
2 Types of Databases

(1) ([8]**Hierarchical database**) : Database where data is represented in a tree-like hierarchical structure.
(2) ([9]**Network database**) : Database where data is represented in a structure similar to a web or mesh.
(3) ([10]**Relational database**) : Database where collected data is organized and managed across multiple tables.

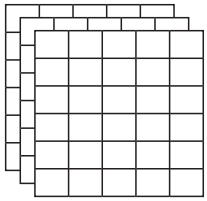
Hierarchical database



Network database



Relational database



(4) ([11]**NoSQL**) : Database management systems other than relational database management systems.

Warm Up

Answer the following questions.

(1) What is the term for information that is organized and stored in a format that makes it easy to access for a specific purpose?

(2) Choose one correct statement regarding data integrity from the options **A** to **D** below, and answer using the letter.

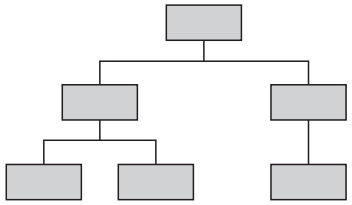
A Function where a database is managed separately from the programs that use it.

B Function that grants access rights to the data being used and limits users.

C Function that prevents duplication, tampering, and unauthorized registration or updating of data.

D Function for backing up and restoring to prepare for data failures.

(3) What is the term for a database with a structure like the one shown on the right?



Explanation

(1) Database

(2) **C**

(3) Data is represented in a tree-like hierarchical structure, therefore hierarchical database

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> Data that is managed and stored in a certain format to make it easier to use large amounts of data is referred to as a ([1]). A system that creates, operates, and manages a ([1]) is called a ([2]), and it has functions such as data consistency, integrity, independence, ([3]), and ([4]).

(2) Choose the correct explanation for each of the terms [1] to [5] related to database management systems from the options **A** to **E**, and answer using the letters.

[1] Data independence [2] Data integrity [3] Data consistency
[4] Data availability [5] Data confidentiality

A Setting access permissions and performing authentication for data.
B Providing a mechanism to ensure there are no inconsistencies in the relationships between data.
C Method where stored data is managed separately according to its use.
D Backing up and taking other measures so that it is easier to recover data after a failure.
E Providing a mechanism to prevent accidental revisions to or unauthorized tampering of data.

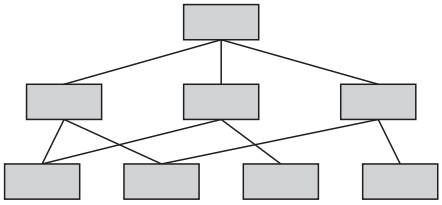
2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

> There are several types of databases, including a ([1]) where collected data is organized and managed across multiple tables. Data that is organized in a hierarchical structure resembling a tree is referred to as a ([2]), while data expressed in a network-like structure is referred to as a ([3]).

A Hierarchical database **B** Relational database
C Network database **D** Object-oriented database

(2) What is the term for a database with a structure like the one shown on the right?



(3) What is the term for database management systems other than relational database management systems?

Exercise

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> A ([1]) refers to data that is organized and stored in a certain format for large-scale use. A ([1]) not only makes data easy to store, but also to search, process, and share. A ([2]) is a system for creating, operating, and managing a ([1]).

(2) Choose one correct statement regarding data availability from the options **A** to **D** below, and answer using the letter.

A Function where a database is managed separately from the programs that use it.
B Function that grants access rights to the data being used and limits users.
C Function that prevents duplication, tampering, and unauthorized registration or updating of data.
D Function for backing up and restoring to prepare for data failures.

(3) For the following statements [1] to [5] regarding the functions of database management systems, mark "○" if the statement is correct, and "×" if it is incorrect.

[1] Data confidentiality involves setting data access rights and implementing authentication.
[2] Data integrity means ensuring that concurrent operations for shared data do not cause inconsistencies.
[3] Data consistency involves not only preventing data duplication but also protecting against data tampering and the registration of fraudulent data.
[4] Data availability involves performing backups, restoration, and recovery to prepare for data failures.
[5] Data independence refers to managing databases separately from the programs that use them.

2 Answer the following questions.

(1) Choose the appropriate database for the following items [1] to [3] from the options **A** to **C** below, and answer using the letters.

[1] A database that represents data in a mesh-like structure.
[2] A database where collected data is organized and managed across multiple tables.
[3] A database where data is represented in a tree-like hierarchical structure.

A Hierarchical database **B** Network database
C Relational database

(2) What is the term for a database with a structure like the one shown on the right?

